Agnibh Dasgupta

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Education

University of Nebraska Omaha, Ph.D., Computing & Information Science *2021 – 2026*
Utah State University, M.S., Computer Science *2018 – 2020*
West Bengal University of Technology, B.Tech., Computer Science *2013 – 2017*

Skills

Research Areas:	Representation Learning, Watermarking, Computer Vision, NLP, Multimodal Learning
Architectures:	Transformers, ViT, CLIP, LLMs, Diffusion Models
Programming:	Python, PyTorch, TensorFlow, Hugging Face Transformers
Distributed:	PyTorch DDP, DeepSpeed, FSDP, Slurm/HPC
Post-Training:	LoRA, SFT, DPO, GRPO, RLHF, RLAIF
Agentic AI:	LangGraph, RAG, Tool Use, Function Calling
Evaluation:	LLM-as-a-Judge, RAGAS, A/B Testing, Human-in-the-Loop
Deployment:	vLLM, Quantization

Awards

CVPR Compute Transparency Champion Award 2026
- Awarded for exemplary reporting of computational resources, methodology, and reproducibility details

Professional Experience

University of Nebraska Omaha and **Utah State University**, Graduate Researcher *2021 – 2026*
Research centered on representation learning and watermarking across vision and language, involving fine-tuning and evaluating LLMs across multiple open-source model families and multimodal vision-language models built on CLIP and vision transformers, evaluated for robustness under adversarial perturbations on large standard datasets. Recent work analyzes internal representations of LLMs for model attribution, using causal intervention to validate findings. Built and deployed an agentic RAG system, optimizing a deployed LLM for inference with vLLM and quantization, and ran a comparative evaluation of post-training alignment methods using a custom human-in-the-loop preference dataset.

Selected Research and Projects

Siosa's Library: Path of Exile Q&A Agent *2026*
- Multi-agent agentic workflow: LangGraph coordinates query expansion, tool selection across multiple data sources
- Self-hosted a pretrained LLM on cloud infrastructure, optimized for inference using vLLM and quantization
- Collected human preference pairs on retrieval quality and used them for post-training to align model behavior

Invariant Features in LLMs: Geometric Characterization and Model Attribution *(under review)* *2026*
- Identified invariant subspaces in 9 open-source LLMs via contrastive sensitivity decomposition
- Validated semantic and nuisance components causally through next-token prediction interventions
- Achieved 90%+ zero-shot model attribution accuracy across base, fine-tuned, and distilled LLMs

TIACam: Text-Anchored Invariant Feature Learning with Auto-Augmentation *CVPR 2026*
- Built a zero-watermarking framework combining CLIP image–text features with a learned augmentation module
- Trained a differentiable auto-augmentor to find camera-like distortions, removing hand-designed augmentation
- Achieved 95%+ watermark recovery on ImageNet and Visual Genome under real-world camera conditions

Watermarking Language Models through Language Models *IEEE TAI 2025*
- Built a model-agnostic watermarking scheme for LLM-generated text without touching model weights
- Tested on 25 open-source LLMs including fine-tuned and distilled variants using LoRA and DeepSpeed
- Achieved 88%+ detection accuracy under paraphrasing, fine-tuning, and adversarial prompt attacks

Robust Image Watermarking based on Cross-Attention and Invariant Domain Learning *CSCI 2023*
- Built a ViT-based cross-attention watermarking system with a self-supervised triplet loss objective
- Trained the encoder to cluster augmented views while separating different images in latent space
- Reached state-of-the-art comparable robustness under compound noise, confirmed via ablation

Perspective Transformation Layer *CSCI 2022*
- Proposed a lightweight learnable layer for perspective-robust recognition, replacing fixed pipelines
- Evaluated on SVHN, Imagenette, and MNIST under viewpoint variation with minimal extra compute
- Outperformed prior state-of-the-art on perspective benchmarks as a drop-in module